

MACHINE LEARNING

END-SEMESTER PROJECT PROPOSAL

Cross-Domain Hallucination Detection via Invariant Hidden-State Probing in LLMs

Advanced Machine Learning Course · Team of 4 · 3-Week Sprint
Submitted: May 24, 2026

DOMAIN	TIMELINE	COMPUTE	TARGET VENUE
NLP / LLMs	3 Weeks	Free Colab T4	ACL / EMNLP 2026

1. EXECUTIVE SUMMARY

Problem: LLM hallucination detectors based on hidden-state probing achieve strong in-domain accuracy but collapse when tested on unseen domains — a critical gap that blocks real-world deployment. **This project builds the first domain-invariant hallucination probe** using contrastive and adversarial training, running entirely on free-tier Colab hardware.

Large Language Models (LLMs) frequently generate confident but factually incorrect outputs — a phenomenon known as hallucination. Recent work (SAPLMA 2023, INSIDE 2024, LLM-Check NeurIPS 2024) shows that internal hidden states encode hallucination signals detectable by lightweight classifiers. However, a 2025 study found these probes fail out-of-distribution (OOD), and a May 2026 paper (PARALLAX) showed most reported benchmark progress is explained by dataset artifacts. This project directly attacks both problems.

2. RESEARCH PROBLEM & NOVELTY

The Core Gap

Existing Method	What It Does	Critical Failure
SAPLMA (2023)	MLP probe on hidden states	OOD accuracy drops ~30%
INSIDE (2024)	Eigenvalue consistency	Needs multiple LLM runs
LLM-Check (2024)	Attention + logit features	Benchmark artifacts
MIND (2024)	Unsupervised hidden-state probe	No cross-domain eval

Our Novel Contribution

- **C1 — Domain-Invariant Probe:** First hallucination probe trained with domain-adversarial objectives to generalize across medical, legal, commonsense, and coding domains without retraining.
- **C2 — Contrastive Representation:** Apply supervised contrastive loss so hallucination representations cluster independently of domain — a novel training paradigm for probing classifiers.
- **C3 — Artifact-Free Benchmark Protocol:** Controlled evaluation that eliminates the benchmark construction artifacts identified by PARALLAX (2026), enabling honest comparison.
- **C4 — Lightweight + Colab-Runnable:** Full pipeline runs on a single T4 GPU (free tier) — democratizing reproducibility for the research community.

3. METHODOLOGY

System Architecture

```
Input Prompt + LLM Answer ↓ Frozen LLM (Mistral-7B 4-bit / GPT-2-Medium) ↓ Hidden States @  
Layers [8, 16, 24, 32] ← Forward hooks ↓ Mean-Pool over sequence →  $\mathbf{h} \in \mathbb{R}^{\text{hidden\_dim}}$   
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ ■ Domain-Adversarial Probe ■ ■ Encoder →  
Hallucination Head ■ ■ ■ Domain Head (reversed) ■ ■ Loss =  $L_{\text{halluc}} + \lambda \cdot L_{\text{domain}}$  ■  
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ ↓ Output: Hallucination Risk Score [0.0 -  
1.0]
```

Training Objective

The probe is trained with a compound loss combining three terms:

- **L_halluc** — Binary cross-entropy for hallucination classification (main objective)
- **L_domain** — Domain adversarial loss via gradient reversal layer (forces domain-invariant representations)
- **L_contrastive** — Supervised contrastive loss pulling same-label samples together across domains
- **L_total = L_halluc + $\lambda \cdot L_{\text{domain}}$ + $\gamma \cdot L_{\text{contrastive}}$** ($\lambda=0.1$, $\gamma=0.05$ from validation)

Datasets

Dataset	Domain	Size	Split
TruthfulQA	General QA	817	Train / Test
HaluEval	Multi-domain	30,000	Train / Test
MedHalt	Medical	8,000	OOD Test
LegalBench	Legal	5,400	OOD Test
WikiBio GPT3	Biography	4,000	OOD Test

4. EXPERIMENTS & EVALUATION PLAN

Baselines to Beat

- SAPLMA — Standard MLP probe on hidden states (Azaria & Mitchell 2023)
- Output-only classifier — BERT trained on LLM text outputs only
- MIND — Unsupervised hidden-state probe (Su et al. 2024)
- LLM-Check — Attention + logit features (NeurIPS 2024)
- Random baseline — 50% for calibration

Expected Results Table

Method	In-Domain F1	Medical OOD	Legal OOD	Avg OOD
Random	50.0	50.0	50.0	50.0
Output Classifier	61.2	54.3	52.1	53.2
SAPLMA	74.8	49.6	47.2	48.4
MIND	72.1	51.3	50.8	51.1
Ours (HalluProbe)	79.4	72.1	70.8	71.5

** Numbers are projected targets; actual results will be reported in final paper.*

Ablation Studies

- Remove domain-adversarial loss → measures its contribution to OOD generalization
- Remove contrastive loss → isolates effect of representation clustering
- Single-layer vs. multi-layer features → which layers matter most
- GPT-2-Medium vs. Mistral-7B-Q4 → probe transferability across model families
- K=1 vs. K=4 domain adversarial heads → sensitivity to domain granularity

5. 21-DAY PROJECT TIMELINE

Phase	Days	Deliverable	Owner
Data Pipeline	1 – 3	5 datasets cleaned, labeled, split	Data Eng.
Baseline Repr.	4 – 5	SAPLMA reproduced, hooks working	ML Lead
Core Model	6 – 9	Adversarial + contrastive probe v1	ML Lead
OOD Eval	10 – 12	Cross-domain benchmark results	Eval Lead
Ablations	13 – 15	Ablation table complete	Full Team
Analysis	16 – 17	Layer heatmaps, attention viz	ML Lead
Demo Build	18	Gradio app live on Colab	Systems
Paper Draft	19 – 20	6-page ACL workshop submission	Full Team
Final Polish	21	GitHub repo, arXiv preprint upload	Full Team

6. TEAM ROLES & RESPONSIBILITIES

Role	Responsibilities
ML Lead	Model architecture, training loop, activation hooks, ablations
Data Engineer	Dataset pipelines, label construction, benchmark protocol design
Evaluation Lead	Metrics, OOD test harness, artifact-free eval framework
Systems / Demo	Gradio demo, GitHub repo, arXiv submission, paper formatting

7. TECHNOLOGY STACK

Category	Tools
Models	GPT-2-Medium, Mistral-7B (4-bit QLoRA via bitsandbytes)
Training	PyTorch, HuggingFace Transformers, TRL, PEFT

Experiment Track	Weights & Biases (free tier), MLflow
Data	HuggingFace Datasets, Pandas, scikit-learn
Demo	Gradio, FastAPI
Compute	Google Colab T4 (free) + Kaggle P100 (free)
Version Control	GitHub + DVC for dataset versioning

8. RISK ASSESSMENT & MITIGATION

Risk	Likelihood	Impact	Mitigation
OOD improvement is marginal	Medium	High	Pre-register hypothesis; ablations still publishable
Colab session timeouts	High	Medium	Checkpoint every 500 steps; use Kaggle as backup
Dataset label noise	Medium	Medium	Manual review of 200 random samples; kappa agreement score
Mistral-7B OOM on T4	Low	Medium	Fall back to GPT-2-XL (1.5B); 4-bit quant as primary
Negative results	Low	Low	Negative result + artifact-free benchmark = still publishable

9. EXPECTED OUTCOMES & SUCCESS METRICS

- **Primary:** Domain-invariant probe achieving >70% F1 on all OOD test domains (vs. ~49% for SAPLMA baseline)
- **Codebase:** Clean GitHub repo with 4 reproducible Colab notebooks, full documentation, and Gradio demo with shareable URL
- **Paper:** 6-page submission to ACL 2026 Workshop on Trustworthy NLP or EMNLP Findings — with arXiv preprint uploaded Day 21
- **Benchmark Protocol:** Artifact-free evaluation framework released as open-source contribution to the community
- **Portfolio Signal:** A complete, end-to-end ML research project demonstrating mechanistic interpretability skills — directly relevant for internships at Anthropic, DeepMind, Meta FAIR, and top university labs

10. KEY REFERENCES

- [1] Azaria & Mitchell (2023). The Internal State of an LLM Knows When It's Lying. EMNLP Findings.
- [2] Chen et al. (2024). INSIDE: LLMs' Internal States Retain the Power of Hallucination Detection. ICLR.
- [3] Sriramanan et al. (2024). LLM-Check: Investigating Detection of Hallucinations in LLMs. NeurIPS.
- [4] Su et al. (2024). MIND: Unsupervised Real-Time Hallucination Detection via Internal States.
- [5] PARALLAX (2026). Separating Genuine Progress from Benchmark Artifacts in Hallucination Detection.
- [6] Kossen et al. (2024). Semantic Entropy Probes for Hallucination Detection.
- [7] Ganin & Lempitsky (2015). Domain-Adversarial Training of Neural Networks. JMLR.
- [8] Khosla et al. (2020). Supervised Contrastive Learning. NeurIPS.